

Richeek Das

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Research Interests

Self Supervised Representation Learning, Robotic Autonomy, Multimodal Sensing, Event-based Vision

Education

University of Pennsylvania

2023 - 2028

PhD in Computer and Information Science

Philadelphia, USA

MS by Research in Computer and Information Science

Advisor: Prof. Pratik Chaudhari

Indian Institute of Technology Bombay

2019 - 2023

Bachelor of Technology with Honors in Computer Science and Engineering

Mumbai, India

Advisor: Prof. Preethi Jyothi

GPA: 9.62/10.0

Thesis: Code-switched Text Modelling for Natural Language Understanding

🏆 **Research Excellence Award for best undergraduate thesis**

Industry Experience

Aurora Innovation, Perception Intern – Multimodal Sensor Fusion and Pretraining

May 2026 - Aug 2026

Managers: Dr. Gary Yang, Dr. Ben Smith

Pittsburgh, PA

- Pioneered the **first self-supervised joint pretraining** and **multimodal fusion** framework for perception using camera and LiDAR – unlocked the use of large-scale unlabelled autonomous trucking data for improving model performance
- First to enable vision transformers in the perception stack **deployed on truck hardware**, with significant accuracy gains

Abacus.AI, Research Intern – Fairer and More Accurate Tabular Models Through NAS 🐙

May 2023 - Jul 2023

Manager: Dr. Samuel Dooley

Remote (Mumbai / San Francisco)

- Proposed a novel multi-objective **Neural Architecture Search** technique to utilize the implicit fairness in model architectures. Proved the existence of inherently fair architectures – surpassing SOTA debiasers. Work accepted at the **NeurIPS Workshop 2023**

Sony AI, Research Intern – Neural Architecture Search Toolkit for Computer Vision 🐙

May 2022 - Jul 2022

Managers: Takuya Narihira, Hsingying Ho

Tokyo, Japan

- Proposed a novel Once-For-All based NAS framework for Semantic Segmentation in NNabla NAS – builds dynamic sub-networks under differing hardware constraints **without any re-training**
- Adapted the method to DeepLabv3+ with dynamic sub-networks – **15% increase in MIoU** and **2× latency reduction**

Academic Research Experience

UPenn, GRASP Lab, Graduate Researcher

Jan 2024 - Ongoing

Advisors: Prof. Pratik Chaudhari, Prof. Kostas Daniilidis, Prof. C.J. Taylor

Philadelphia, PA

Worked on robotics, perception, representation learning and simulation research.

- **Representation Learning for Event-based Vision** 🐙 – theory and efficient algorithms to build representations of event camera data. SOTA performance on optical flow, segmentation and depth on 3 platforms (car, quadruped, drone), different lightings and environments (indoors, outdoors, off-road). Accepted at **Science Robotics 2026**
- **Real-time Multirotor Simulation Software** 🐙 – very fast simulation environment for multirotors with IMUs, RGB, event, depth and semantic cameras. Controllers, RL, planning, perception and HITL support. CUDA kernels for 10–250× faster event simulation (~2700 FPS on RTX4090). **Oral at CVPRW 2026**

- **Quadrotor Ecosystem for Event-based Autonomy** – compact Jetson Orin NX quadrotor with an event camera and a full ROS 2 autonomy stack—runs new onboard event-based VIO, RL and monocular depth-based navigation, planning, and reactive obstacle dodging using monocular events + IMU. In preparation for **IEEE T-RO Journal**

UPenn, WAVES Lab, Extracting Scene Material Properties from Radar Signals 🦾

Aug 2023 - Jan 2024

Advisor: Prof. Mingmin Zhao

Philadelphia, PA

- Built a device with synchronized iPad **LiDAR**, TI mmWave **radar**, camera and a Jetson to scan indoor scenes with diverse materials. Built a differentiable radar simulator to model wave propagation, antenna patterns and **ray tracing**, and a new SSL framework to extract material properties using the physics of wave-material interactions

Google India + IIT Bombay, CSALT, Code-Switched Natural Language Understanding 🦾

Jul 2022 - Jan 2023

Advisor: Prof. Preethi Jyothi

Mumbai, India

- Implemented intelligent masking strategies for MLM **pretraining** on code-switched text, improving downstream Question Answering and Sentiment Analysis. Built a framework adapting translation models for low-resource conditioned **language generation** – **ACL 2023 Outstanding Paper Award**

IIT Bombay, Weighted LASSO under Multiplicative LogNormal Noise 🦾

Jan 2022 - Feb 2023

Advisor: Prof. Ajit Rajwade

Mumbai, India

- Designed a novel Weighted LASSO for sparse **signal recovery** under multiplicative LogNormal noise. Theoretically and empirically justified pooled RT-PCR testing across hospitals, accepted at **Signal Processing Journal 2024**

University of Vienna, Neuroinformatics Lab, Kernels for Nonlinear Causal Clustering 🦾

May 2021 - Oct 2021

Advisor: Prof. Moritz Grosse-Wentrup

Vienna, Austria

- Implemented, simulated and evaluated a distance covariance-based **kernel** that clusters samples by their parent **causal structures** in heterogeneous populations, accepted at **CLear 2022**

TU Delft, Delft AI Labs, Belief Elicitation on the Impact of AI Systems 🦾

May 2021 - Oct 2021

Advisor: Prof. Ujwal Gadiraju

Delft, Netherlands

- Implemented a Binarized Scoring Rule criterion to elicit user trust in AI decision support for **algorithmic bargaining**. Built a DRF, Angular, PostgreSQL app hosting **2700+ crowdsourced submissions**, accepted at **ACM CHI 2022**

IIT Gandhinagar, CVIG Lab, Feature Gating for Deeper Graph Convolution Networks

Dec 2020 - Jun 2021

Advisor: Prof. Shanmuganathan Raman

Gandhinagar, India

- Introduced feature gating and formulated mechanisms to award importance scores to graph nodes. Tackled over smoothing in **deep GCN models** – SOTA **semi-supervised** node classification, accepted at **IEEE ICASSP 2022**

Journal Publications

Fast Feature Field (F^3): A Predictive Representation of Events 🦾

Richeek Das, Kostas Daniilidis, Pratik Chaudhari

Science Robotics 2026 (Patent Pending)

Neurofly: An Open-Source High-Performance Quadrotor Ecosystem for Event-based Autonomy

In preparation for **IEEE T-RO**

Performance Bounds for LASSO under Multiplicative LogNormal Noise: Applications to Pooled RT-PCR Testing 🦾

Richeek Das, Aaron Jerry Ninan, Adithya Bhaskar, Ajit Rajwade

Signal Processing Journal (Elsevier) 2024

Conference Publications

Pretraining Techniques for Improved Code-switched Natural Language Understanding

Richeek Das, Sahasra Ranjan, Shreya Pathak, Preethi Jyothi

Association for Computational Linguistics (**ACL**) 2023 🏆 **Outstanding Paper Award**, ~ 1% of accepted papers

A Distance Covariance-based Kernel for Nonlinear Causal Clustering in Heterogeneous Populations 🦋

Alex Markham, **Richeek Das**, Moritz Grosse-Wentrup

Conference on Causal Learning and Reasoning (**CLearR**) 2022

For What It's Worth: Humans Overwrite Their Economic Self-interest to Avoid Bargaining With AI Systems

Alexander Erlei, **Richeek Das**, Lukas Meub, Avishek Anand, Ujwal Gadiraju

ACM Conference on Human Factors in Computing Systems (**CHI**) 2022

Exploring Deeper Graph Convolutions For Semi-Supervised Node Classification

Ashish Tiwari, **Richeek Das**, Shanmuganathan Raman

IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**) 2022

Workshop Publications

Neurosim: A Fast Simulator for Neuromorphic Robot Perception 🦋

Richeek Das, Pratik Chaudhari

CVPR Workshop on the Science of Simulation (SoS) 2026 🏆 **Oral Presentation**

Fairer and More Accurate Tabular Models Through NAS 🦋

Richeek Das, Samuel Dooley

NeurIPS Workshop on Algorithmic Fairness through the Lens of Time 2023

Selected Projects

Computational Imaging and Compressed Sensing

- **Video from a Single Exposure Coded Snapshot** 🦋 – compressive temporal imaging in MATLAB, recovering frame sequences from one snapshot Spring 2021
- **Compressed Sensing in Tomography** 🦋 – brain MRI reconstruction from few measurements Spring 2021

Computer Systems and Infra

- **Compiler for a C-like Language** – parser in lex and yacc with type inference and AST to machine code Spring 2022
- **Extending xv6 Operating System** – syscalls for demand paging and custom fork, thread synchronization, semaphores and a disk-emulated filesystem in C Autumn 2021
- **Branch Predictors for Trace-based Simulators** – TAGE and L-TAGE in ChampSim, compared against Bi-modal and Hashed Perceptron on MPKI and history length Autumn 2021
- **Cluster Monitoring and Alert System** – DRF, Angular and InfluxDB app with telegraf agents for real-time cluster profiling and low-resource alerts Spring 2022

Review Experience

- Conference on Robot Learning (CoRL) - 2026
- Science Robotics - 2026
- Signal Processing Journal (Elsevier) - 2026
- Natural Language Processing Journal (Springer) - 2024
- CVPR Workshop on Multimodal Algorithmic Reasoning (MAR) - 2026

Teaching Experience

Graduate Teaching Assistant

University of Pennsylvania

- Learning in Robotics, *Prof. Pratik Chaudhari*

Spring 2026

Undergraduate Teaching Assistant

IIT Bombay

- Introduction to Machine Learning, *Prof. Biplab Banerjee, Prof. Manjesh Hanawal*
- Teaching Practicum for TAs, *Dept. of Computer Science and Engineering*
- Computer Networks, *Prof. Vinay Ribeiro*
- Software Systems Lab ([Excellence in CSE TAs](#) Award), *Prof. Amitabha Sanyal*

Spring 2023
Autumn 2022, Spring 2023
Autumn 2022
Autumn 2021

- Engineering Drawing, *Prof. Atul Sharma*
- Computer Programming and Utilization, *Prof. Kameswari Chebrolu*

Summer 2021
Spring 2021

Selected Awards and Honors

- Received the **ACL 2023 Outstanding Paper Award**, awarded to the **top 1%** of the accepted papers 2023
- Received the **Paul S. Darnell** named CIS PhD fellowship at UPenn 2023
- Received the **Thomas Dooie, Class of 1974 Research Excellence Award** for B.Tech Thesis 2023
- Among the select few undergrads to attend the **Google Research Week** hosted in Bangalore, India 2023
- Secured a rank of **497** in JEE Main and **544** in JEE Advanced among **1.2 million** candidates 2019
- Received the **INSPIRE** scholarship, awarded to the **top 1%** of **80k+** students in the **ISC** exam 2019
- Received the **KVPY Fellowship** for securing **77th** rank out of **150k+** candidates in all of India 2018
- Secured **4th rank in India** in **ICSE** out of **180k+** candidates 2017